

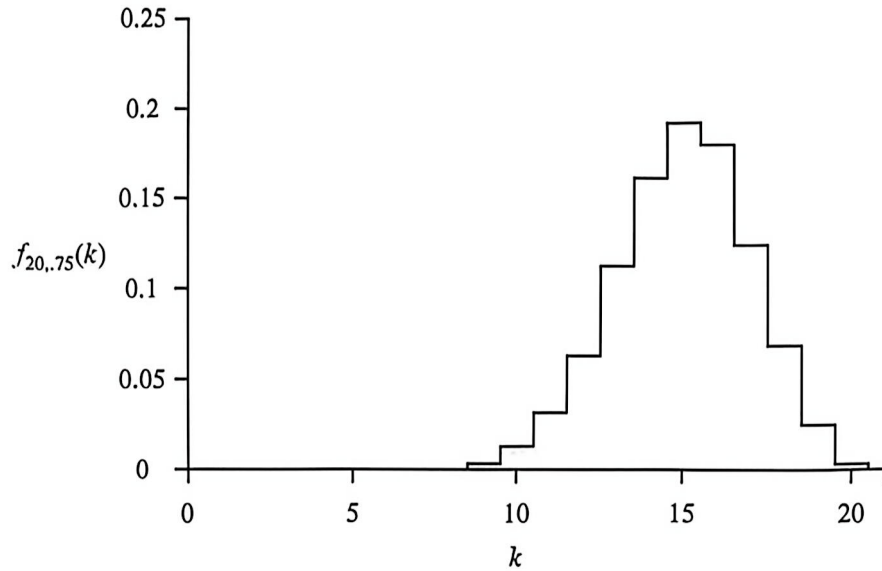


Figure 18.5 The pdf for the general binomial distribution $f_{n,p}(k)$ for $n = 20$ and $p = .75$.

18.4 Great Expectations

The *expectation* or *expected value* of a random variable is a single number that reveals a lot about the behavior of the variable. The expectation of a random variable is also known as its *mean* or *average*. For example, the first thing you typically want to know when you see your grade on an exam is the average score of the class. This average score turns out to be precisely the expectation of the random variable equal to the score of a random student.

More precisely, the expectation of a random variable is its “average” value when each value is weighted according to its probability. Formally, the expected value of a random variable is defined as follows:

Definition 18.4.1. If R is a random variable defined on a sample space S , then the expectation of R is

$$\text{Ex}[R] ::= \sum_{\omega \in S} R(\omega) \Pr[\omega]. \quad (18.2)$$

Let’s work through some examples.